

Chemistry Lab C - Chemistry Lab - Monta Vista Science Olympiad (Div. C) - 11-14-2025

Student Answer Sheet (/monta-vista/SM/AnswerSheet?tid=000GCR)

Instructions (shown before students start the test)

Introduction (shown after students start the test)

Equilibrium

1. (1.00 pts) A 0.0500 mol sample of $\text{PCl}_5(\text{g})$ is placed in a 2.00 L container and allowed to reach equilibrium at 250. °C:

According to the reaction $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$

At equilibrium, the total pressure is 2.12atm

Write the Kp equation for the reaction.

- ☐ A) $K_p = p\text{PCl}_5$
- ☐ B) $K_p = p\text{PCl}_3 * p\text{Cl}_2$
- ☐ C) $K_p = p\text{PCl}_3 * p\text{Cl}_2 / p\text{PCl}_5$
- ☐ D) $K_p = p\text{PCl}_5 * p\text{Cl}_2 / p\text{PCl}_3$

2. (1.00 pts) Calculate the equilibrium partial pressures for all species.

- ☐ A) $p\text{PCl}_5$: 1.08 atm
 $p\text{Cl}_2$: 0.520 atm
 $p\text{PCl}_3$: 0.520 atm
- ☐ B) $p\text{PCl}_5$: 0.0283 atm
 $p\text{Cl}_2$: 1.04 atm
 $p\text{PCl}_3$: 1.04 atm
- ☐ C) $p\text{PCl}_5$: 0.560 atm
 $p\text{Cl}_2$: 0.780 atm
 $p\text{PCl}_3$: 0.780 atm
- ☐ D) $p\text{PCl}_5$: 0.340 atm
 $p\text{Cl}_2$: 0.890 atm
 $p\text{PCl}_3$: 0.890 atm

3. (1.00 pts) What is the Kp of the reaction?

- ☐ A) 1.80
- ☐ B) 1.93
- ☐ C) 0.98
- ☐ D) 1.34

4. (1.00 pts) Assume at 500°C, the K_p of the reaction is 2.00, what would the K_c be?

- ☐ A) 1.48
- ☐ B) 2.00
- ☐ C) 0.0315
- ☐ D) 0.103

5. (1.00 pts) For the equilibrium $\text{Co}(\text{H}_2\text{O})_6^{2+}(\text{aq}) + 4\text{Cl}^-(\text{aq}) \rightleftharpoons \text{CoCl}_4^{2-}(\text{aq}) + 6\text{H}_2\text{O}(\text{l})$. How Does the equilibrium shift if $\text{NaCl}(\text{aq})$ is added?

- ☐ A) left
- ☐ B) right
- ☐ C) none

6. (1.00 pts) If water is added?

- ☐ A) left
- ☐ B) right
- ☐ C) none

7. (1.00 pts) If the solution is heated?

- ☐ A) left
- ☐ B) right
- ☐ C) none

8. (3.00 pts) What analytical technique could you use to confirm these results?

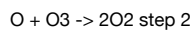
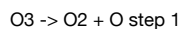
9. (2.00 pts) Design a buffer solution with a pH of 4.76 using acetic acid ($K_a = 1.80 \times 10^{-5}$). What ratio $[\text{A}^-]/[\text{HA}]$ is required?

10. (2.00 pts) If you have 0.10 M acetic acid, how much sodium acetate (in grams) should be added to 100 mL of this solution to achieve the desired pH?

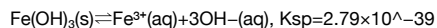
11. (2.00 pts) Briefly describe how you would verify the pH experimentally.

12. (2.00 pts) Using equilibrium, explain why the solubility of Ag^+ decreases as NH_3 is added, then increases as excess NH_3 is increased. 1-2 sentences.

13. (2.00 pts) Find the reaction rate equation for the given reaction mechanism:



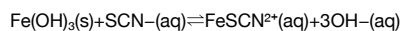
14. (1.00 pts) Given:



and



Calculate the overall equilibrium constant for:



- ☐ A) $K_{\text{eq}} = 1.10 \times 10^3$
- ☐ B) $K_{\text{eq}} = 3.94 \times 10^{41}$
- ☐ C) $K_{\text{eq}} = 2.54 \times 10^{-42}$
- ☐ D) $K_{\text{eq}} = 3.07 \times 10^{-36}$

15. (2.00 pts) Briefly describe how addition of SCN^- would affect $\text{Fe}(\text{OH})_3$ solubility. 1-2 sentences.

16. (1.00 pts) The equilibrium constant K_c for a reaction depends on

- ☐ A) Concentrations of reactants and products
- ☐ B) Temperature
- ☐ C) Pressure
- ☐ D) Catalyst Concentration

17. (8.00 pts)

Complex Equilibrium System

Given K_{a1} and K_{a2} of H_2S is 9.6×10^{-8} and 1.3×10^{-14} , Calculate the molar solubility of CuS ($K_{sp} = 8.0 \times 10^{-37}$) in a solution in which the $[H_3O^+]$ is (a) $2.0 \times 10^{-1} M$; (b) $2.0 \times 10^{-4} M$

(Hint: You will need mass-balance, K_{a1} , K_{a2} , and K_{sp} equations)

Kinetics

18. (1.00 pts) Write the balanced equation for $N_2O_5(g) \rightarrow NO_2(g) + O_2(g)$

- ☐ A) $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$
- ☐ B) $3N_2O_5(g) \rightarrow 6NO_2(g) + 2O_2(g)$
- ☐ C) $N_2O_5(g) \rightarrow 2NO_2(g) + 2O_2(g)$
- ☐ D) $4N_2O_5(g) \rightarrow 6NO_2(g) + 2O_2(g)$

19. (1.00 pts) [N₂O₅] (M) Initial Rate (M/s)

0.100 2.8 × 10⁻⁴

0.050 1.4 × 10⁻⁴

0.025 7.0 × 10⁻⁵

Determine the order of the reaction

- ☐ A) zero
- ☐ B) first
- ☐ C) second
- ☐ D) third

20. (1.50 pts) How does the half-life increase/decrease as the reaction proceeds?

- ☐ A) increase
- ☐ B) decrease
- ☐ C) stays the same

21. (1.00 pts) calculate the rate constant for the reaction

- ☐ A) $k = 2.8 \times 10^{-2} \text{ s}^{-1}$.
- ☐ B) $k = 2.8 \times 10^{-3} \text{ s}^{-1}$.
- ☐ C) $k = 2.8 \times 10^{-1} \text{ s}^{-1}$.
- ☐ D) $k = 2.8 \times 10^{-2} \text{ M}^{-1} \text{ s}^{-1}$.

22. (1.00 pts) If [N₂O₅] = 0.030 M, what is the rate?

- ☐ A) $8.4 \times 10^{-5} \text{ M} \cdot \text{s}^{-1}$
- ☐ B) $2.5 \times 10^{-5} \text{ M} \cdot \text{s}^{-1}$
- ☐ C) $3.0 \times 10^{-2} \text{ M} \cdot \text{s}^{-1}$
- ☐ D) $9.3 \times 10^{-1} \text{ M} \cdot \text{s}^{-1}$

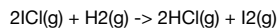
23. (1.00 pts)

H₂O₂(aq) decomposes on its own, but the addition of a small amount of Fe³⁺(aq) will greatly increase the rate that equilibrium is reached. What is the best description of Fe³⁺(aq)?

- ☐ A) a transition state
- ☐ B) reaction intermediate

- ☐ C) heterogenous catalyst
- ☐ D) homogenous catalyst

24. (1.50 pts) The rate constant k for the reaction shown is $1.63 \times 10^{-6} \text{ M}^{-1}\text{s}^{-1}$. What is the overall order of the reaction?



- ☐ A) zero
- ☐ B) first
- ☐ C) second
- ☐ D) third

25. (3.00 pts) What is the most likely order of the decomposition of nitrogen dioxide on a platinum surface?

- ☐ A) zero
- ☐ B) first
- ☐ C) second
- ☐ D) third

26. (1.00 pts) Bismuth-212 undergoes both alpha decay (35.9%) and beta decay (64.1%) with an overall half-life of 60.6 minutes. What is its half-life for alpha decay?

- ☐ A) 21.8min
- ☐ B) 38.8 min
- ☐ C) 94.7 min
- ☐ D) 169 min

27. (1.00 pts) The rate of a chemical reaction generally increases with temperature because:

- ☐ A) A) Reactant molecules have greater mass.
- ☐ B) B) Activation energy decreases with temperature.
- ☐ C) C) More molecules have sufficient energy to overcome the activation energy barrier.

- ☐ D) D) The rate constant becomes independent of temperature.

28. (1.00 pts) Which of the following will not change the rate of a reaction?

- ☐ A) Increasing temperature
- ☐ B) Increasing concentration of reactants
- ☐ C) Adding a catalyst
- ☐ D) Increasing the total pressure of an inert gas in a liquid-phase reaction

29. (1.00 pts) For a second-order reaction, which plot gives a straight line?

- ☐ A) $[A]$ vs. t
- ☐ B) $1/[A]$ vs. t
- ☐ C) $\ln[A]$ vs. t
- ☐ D) rate vs. $[A]$

30. (1.00 pts) A catalyst increases the reaction rate by:

- ☐ A) Increasing the activation energy.
- ☐ B) Providing an alternate pathway with a lower activation energy.
- ☐ C) Increasing the temperature.
- ☐ D) Changing the equilibrium constant.

31. (1.00 pts) The half-life of a first-order reaction depends on:

- ☐ A) Initial concentration
- ☐ B) Temperature
- ☐ C) Catalyst only
- ☐ D) It is constant for all temperatures

32. (1.00 pts) The slowest step in a reaction mechanism is called

- ☐ A) Chain-initiating step
- ☐ B) Chain-terminating step

- ☐ C) Rate-determining step
- ☐ D) Slow equilibrium step

33. (1.00 pts) The Arrhenius equation relates rate constant k to:

- ☐ A) Activation energy and temperature
- ☐ B) Reaction order
- ☐ C) Reactant concentration
- ☐ D) Catalytic efficiency

34. (1.00 pts) If a reaction has a high activation energy, it will

- ☐ A) Vary more with temperature
- ☐ B) Not depend on temperature
- ☐ C) Have a low rate constant
- ☐ D) Proceed very fast

More Equilibrium

35. (1.00 pts) When a chemical system at equilibrium is disturbed, it will shift to oppose the disturbance. This statement is known as

- ☐ A) Hess's Law
- ☐ B) Le Châtelier's Principle
- ☐ C) Boyle's Law
- ☐ D) Arrhenius law

36. (1.00 pts) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$

- ☐ A) The forward reaction
- ☐ B) The reverse reaction
- ☐ C) No change
- ☐ D) Depends on temperature

37. (1.00 pts) At equilibrium, which statement is true

- ☐ A) The rate of the forward reaction is greater than the reverse rate.
- ☐ B) The concentrations of all species are equal.
- ☐ C) The forward and reverse rates are equal.
- ☐ D) The equilibrium constant equals one.

38. (1.00 pts) For the reaction $\text{A} \rightleftharpoons \text{B}$, $K_c = 10$. This means:

- ☐ A) $[A] > [B]$ at equilibrium
- ☐ B) $[B] > [A]$ at equilibrium
- ☐ C) The reaction is not reversible
- ☐ D) The system is at dynamic equilibrium

39. (1.00 pts) If $Q > K$, the reaction will

- ☐ A) Proceed forward
- ☐ B) Proceed in reverse
- ☐ C) Be at equilibrium
- ☐ D) Stop completely

40. (1.00 pts) Adding more solid reactant to a heterogeneous equilibrium

- ☐ A) Shifts equilibrium right
- ☐ B) Shifts equilibrium left
- ☐ C) Has no effect on equilibrium position
- ☐ D) Changes K value

41. (1.00 pts) $\text{CO(g)} + \text{H}_2\text{O(g)} \rightleftharpoons \text{CO}_2\text{(g)} + \text{H}_2\text{(g)}$, is an endothermic reaction. Which change will increase the amount of H_2 formed?

- ☐ A) Removing CO_2
- ☐ B) Adding CO_2
- ☐ C) Adding H_2
- ☐ D) Decreasing temperature

42. (1.00 pts) The value of K_p for a gaseous reaction at constant temperature

- ☐ A) Changes when total pressure changes
- ☐ B) Changes when temperature changes
- ☐ C) Changes when catalyst is added
- ☐ D) Never changes

43. (1.00 pts) Which statement correctly describes a catalyst in an equilibrium system

- ☐ A) It increases K_{eq}
- ☐ B) It decreases activation energy for the forward reaction only
- ☐ C) It speeds up both forward and reverse reactions equally
- ☐ D) It shifts the equilibrium position

44. (0.00 pts) Tiebreaker: What is the K_p of the reaction $2\text{NO(g)} + \text{Cl}_2\text{(g)} \rightarrow 2\text{NOCl(g)}$ at 375K

